2026 Final Year Projects

Detailed System Design Document

Taxila Learning Hub

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Group Number: 17

RAKGETSE PI 202204262

KEKANA PL 240064443

NETSHANDAMA PL 202225805

MOGORU PM 202105210

MUTHALA PW 202343544

LUSUNZI R 240021616

MASHABNA NB 201820548

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Signature: _____

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1. DOCUMENT OVERVIEW

This document has been developed by Ragketse PI , Kekana PL , Netshandama PL , Mogoru PM , Muthala PW, Lusunzi R and Mashaba MB for Texila learning hub for KAEMO TECHNOLOGIES . This document was developed from the project proposal, SRS document, SDLC document and is intended to satisfy all the customer requirements , objectives and expectations

1.1 SCOPE

This document provides the detailed system design for the School Digital Platform to be developed . The system consists of a web application, mobile application(Android 8.0 + & iOS 12+), backend server, centralized database, and an Artificial Intelligence module. The AI module will be used to predict student academic performance and classify students into risk categories using Logistic Regression . This document describes the system architecture, software design, database design, system interfaces, and system performance requirements.

1.2 AUDIENCE

1.**Project Supervisors** :

Require strong knowledge of software engineering, UML, database design, and machine learning concepts.

2.**Development Team (Group 17)**:

Students should be skilled in web, mobile, backend development, and Python for AI.

3.**School Administrators and Teachers**:

Need basic computer literacy and familiarity with school administrative processes.

4.**Learners and Parents:**

Require only basic digital literacy and smartphone usage.

5.**Testers**:

Require understanding of system architecture, data flows, interfaces, and non-functional requirements to develop test cases and validate the system

Assumptions:

-All users have access to internet-enabled devices (smartphones or computers).

-End users possess basic familiarity with mobile apps and web platforms.

-School Administrators can provide anonymized academic data(marks , attendance , homework) for AI model training and testing.

-AI predictions will serve as teacher decision-support tools only.

1.3 RELATED DOCUMENTATION

The following documents are related to this System Design Document:

- Project Proposal Document
- Software Requirements Specification (SRS)
- Software Development Life Cycle (SDLC) Report
- UML Diagrams (Use Case Diagram, System Architecture Diagram, ERD)

1.4 DOCUMENT CONVENTIONS

This document uses Unified Modelling Language (UML) diagrams such as Use Case Diagrams, System Architecture Diagrams, and Entity Relationship Diagrams (ERD) to represent system design and architecture.

2 SYSTEM OVERVIEW

2.1 DESCRIPTION

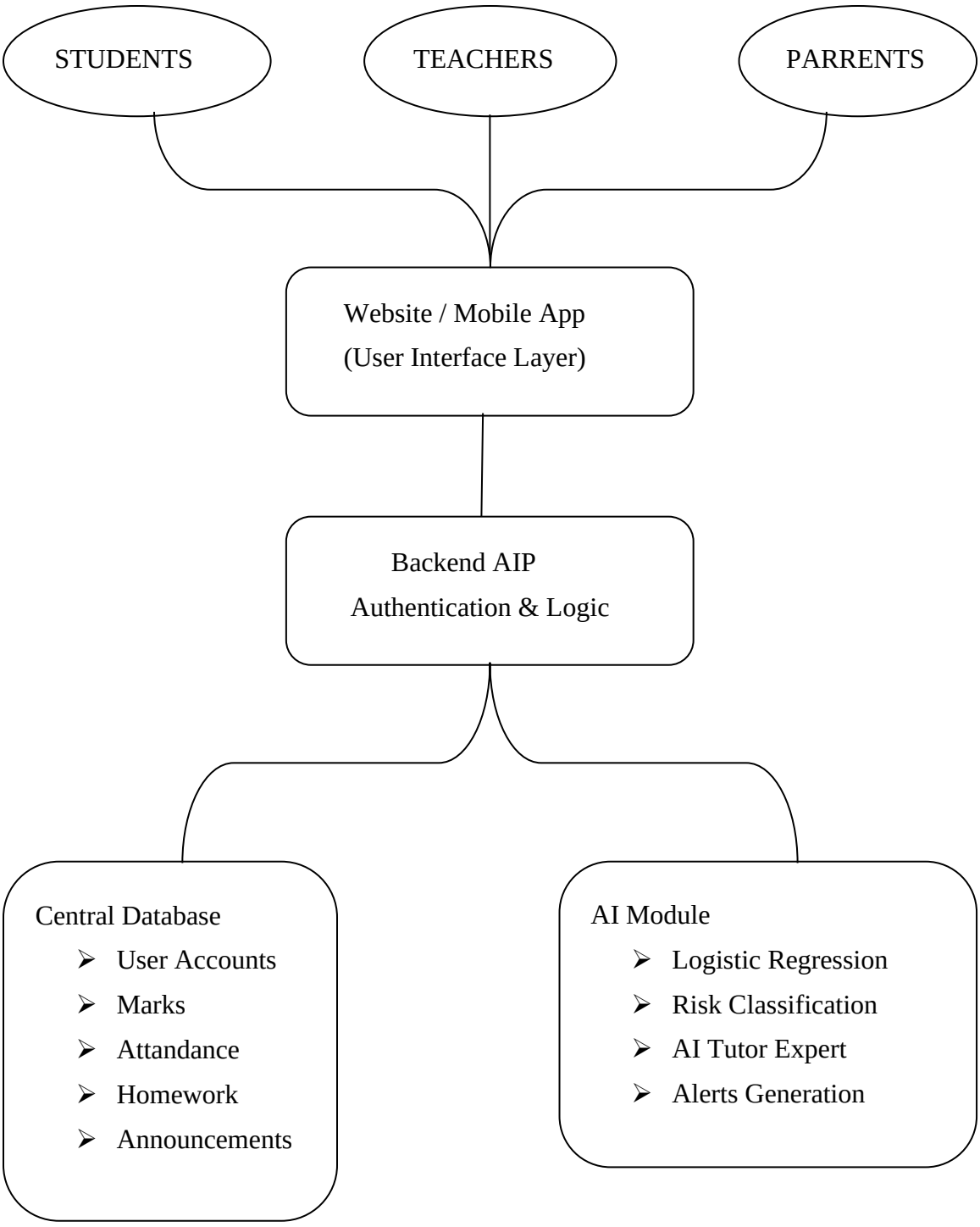
The School Digital Platform is an integrated system designed to improve communication between schools, students, parents, and teachers. The system allows users to access academic information, homework, announcements, attendance records, and student performance reports through a website and mobile application. The system includes an Artificial Intelligence module that analyses student academic data and predicts whether a student is at risk of failing. The system will also include an AI Tutor Expert to assist students with academic support

2.2 SYSTEM ARCHITECTURE

The system follows a three-tier architecture consisting of:

- Presentation Layer (Website and Mobile Application)
- Application Layer (Backend Server and API)
- Data Layer (Database and AI Module)

Users interact with the system through the website or mobile application. The frontend communicates with the backend API, which processes data and connects to the central database and AI module.


STUDENTS
TEACHERS
PARRENTS
Website / Mobile App
(User Interface Layer)
Backend AIP
Authentication & Logic
Central Database
User Accounts
Marks
Attandance
Homework
Announcements
AI Module
Logistic Regression
Risk Classification
AI Tutor Expert
Alerts Generation

2.3 SOFTWARE ARCHITECTURE

The software architecture consists of the following components:

- Frontend: Developed using HTML, CSS, JavaScript
- Mobile Application: Developed using Flutter or React Native
- Backend Server: Developed using Node.js or Python
- Database: MySQL or PostgreSQL
- AI Module: Python with Scikit-learn (Logistic Regression)

2.4 HARDWARE ARCHITECTURES

The system will run on cloud servers and will be accessible through computers, tablets, and smartphones connected to the internet.

3. HARDWARE DESIGN

3.1 HARDWARE COMPONENTS

The system requires the following hardware:

- Web server
- Database server
- User devices (computers, tablets, smartphones)
- Internet connection

3.2 COMPUTER SYSTEMS

The system will run on cloud-based servers and can be accessed using Windows, Android, or iOS devices.

❖ **Peripherals**

Peripherals include user input devices such as keyboard, mouse, and touchscreen devices.

❖ **Networks**

The system will operate over the internet using standard network protocols such as HTTP and HTTPS.

❖ **Hardware Integration**

The frontend devices connect to the backend server through the internet. The backend server communicates with the database server and AI module for data storage and processing.

❖ **PERIPHERALS**

Peripherals include user input devices such as keyboard, mouse, and touchscreen devices.

- **NETWORKS**

  The frontend devices connect to the backend server through the internet. The backend server communicates with the database server and AI module for data storage and processing.

PROJECT SPECIFIC HARDWARE ITEMS (E.G. SENSORS, TRANSDUCERS, ROBOTICS, ENCLOSURE DESIGN)

- **HARDWARE INTEGRATION**

  The frontend devices connect to the backend server through the internet. The backend server communicates with the database server and AI module for data storage and processing.

4. SOFTWARE DESIGN

4.1 SOFTWARE PACKAGES

The system will consist of the following modules:

- **Software Module 1: User Management Module**

Purpose: Handles user registration, login, authentication, and role-based access control.
Users: Students, Teachers, Parents, Administrators
Inputs: Username, Password, User Role
Outputs: Access to system dashboard

- **Software Module 2: Academic Records Module**

Purpose: Stores and manages marks, attendance, and homework records.
Inputs: Student marks, attendance records, homework completion
Outputs: Student performance reports

- **Software Module 3: AI Prediction Module**

Purpose: Predicts whether a student is at risk of failing.
Algorithm Used: Logistic Regression
Inputs: Marks, Attendance, Homework Completion
Outputs: Risk Level (Low Risk, Medium Risk, High Risk)

- **Software Module 4: AI Tutor Expert Module**

Purpose: Provides academic assistance and explanations to students.
Inputs: Student questions
Outputs: AI-generated explanations and learning guidance

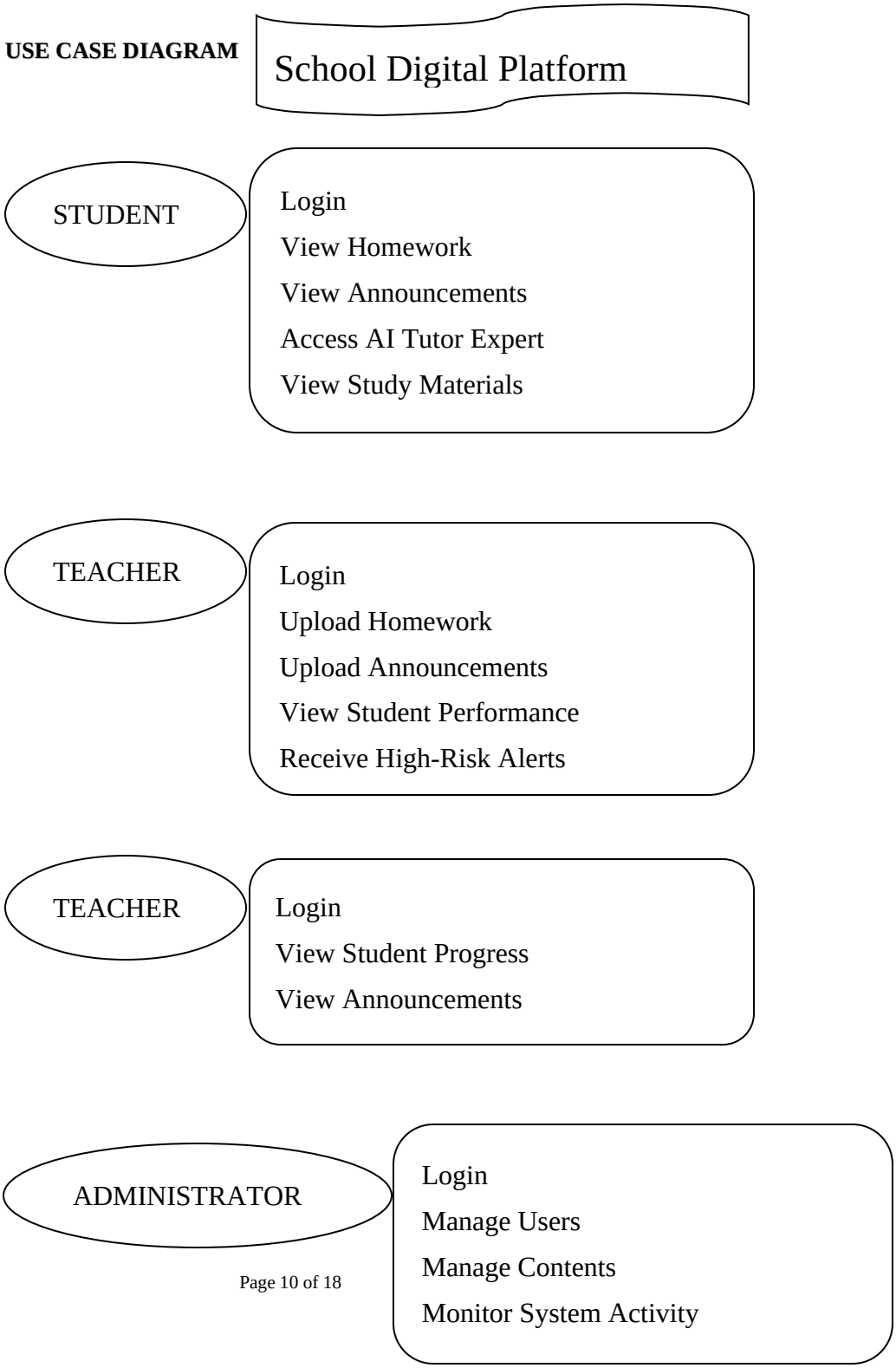
- **Software Module 5: Notification Module**

Purpose: Sends alerts to teachers when a student is identified as High Risk.
Outputs: Email notifications and system alerts

4.2 SOFTWARE INTEGRATION

The website and mobile application communicate with the backend server through APIs. The backend server communicates with the database and AI module to process data and generate predictions.

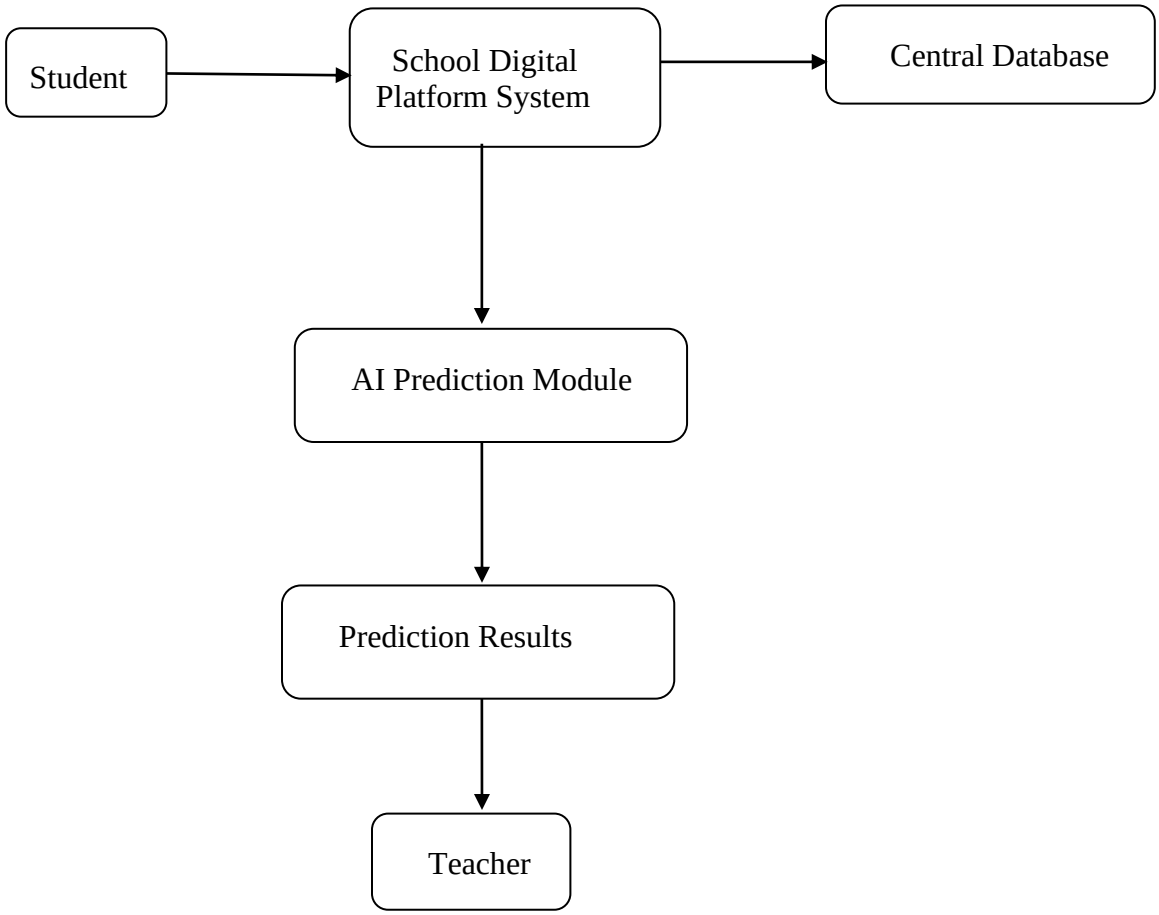
4.3 USE CASE DIAGRAM



The use case diagram illustrates the interaction between the main users and the School Digital Platform. Students, teachers, parents, and administrators each perform specific functions within the system according to their roles.

5. DATA AND DATABASE / FILES

5.1 DATA FLOW DIAGRAMS



The Data Flow Diagram illustrates how student data moves through the School Digital Platform. Student information is entered into the system, stored in the database,

processed by the AI prediction module, and then returned as prediction results to the teacher for academic intervention.

5.2 DATABASE DESIGN (ERD)

The School Digital Platform will use a relational database to store and manage system data such as user information, student academic records, attendance, homework, announcements, Artificial Intelligence predictions, and notifications. The database will be implemented using MySQL or PostgreSQL and will support secure storage and retrieval of data.

Database Tables

The main database tables include:

- Users
- Students
- Teachers
- Parents
- Marks
- Attendance
- Homework
- Homework_Submissions
- Announcements
- AI_Predictions
- Notifications

Each table will have a primary key to uniquely identify records and foreign keys to maintain relationships between tables.

Entity Relationships

The relationships between entities are as follows:

- One User can be a Student, Teacher, Parent, or Administrator.
- One Parent can have many Students.

- One Student can have many Marks records.
- One Student can have many Attendance records.
- One Student can have many Homework Submissions.
- One Student can have many AI Prediction records.
- One Teacher can create many Homework records.
- One Teacher can post many Announcements.
- One Homework can have many Homework Submissions.
- One User can receive many Notifications.

Schema and Query Language

The database schema will be implemented using a relational database model. SQL (Structured Query Language) will be used to create tables, insert data, update records, delete records, and retrieve information from the database.

Keys and Indices

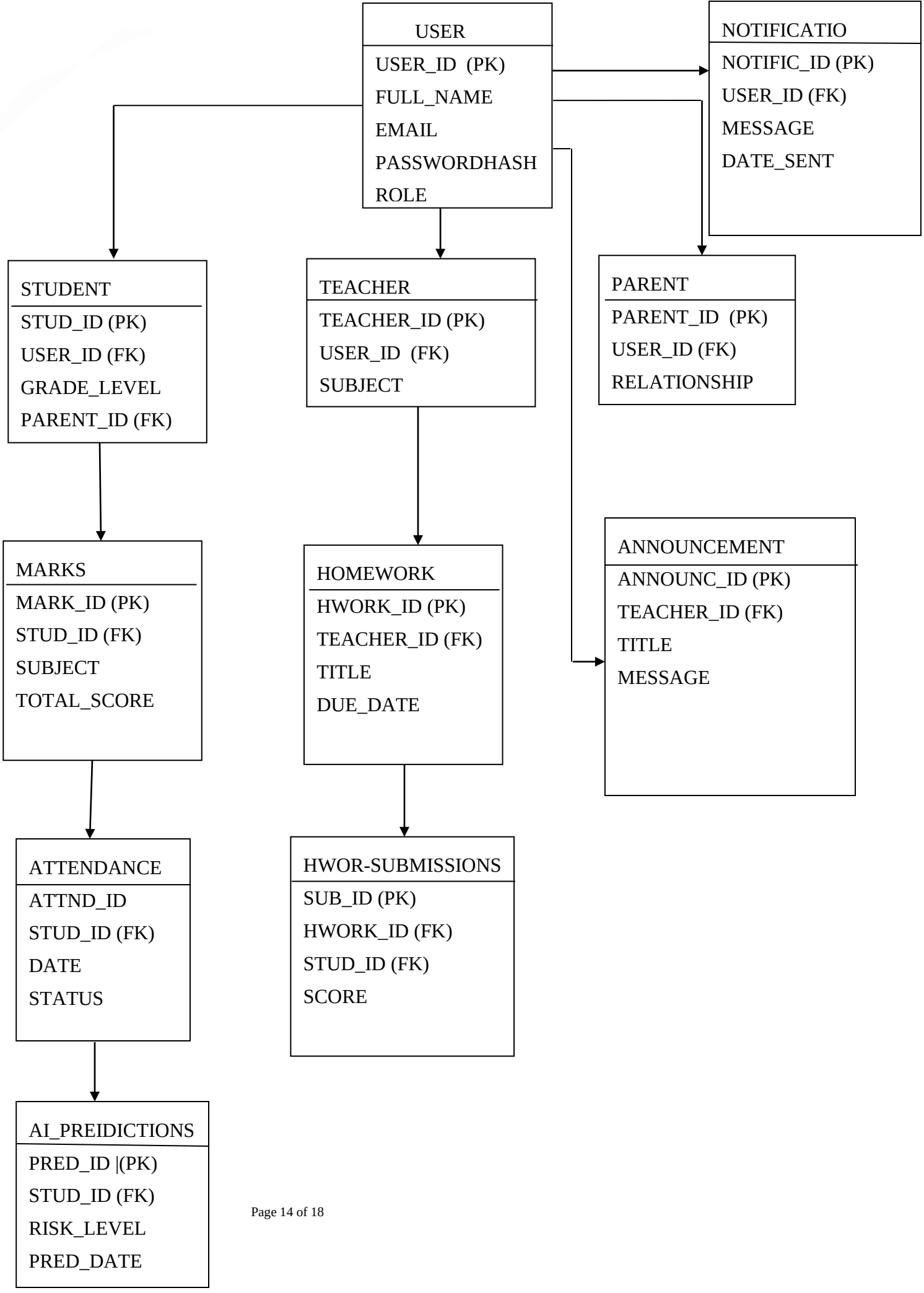
Primary keys will be used to uniquely identify each record in a table, and foreign keys will be used to create relationships between tables. Indexes will be created on commonly searched fields such as StudentID, Email, and Date to improve database performance.

Data Management Functions

The database will support the following functions:

- User registration and authentication
- Storage of academic records
- Attendance tracking
- Homework management
- Announcement management
- AI prediction storage
- Notification management
- Database backup and recovery

An Entity Relationship Diagram (ERD) will be used to visually represent the database structure and relationships between entities.


USER
USER_ID (PK)
FULL_NAME
EMAIL
PASSWORDHASH
ROLE
NOTIFICATIO
NOTIFIC_ID (PK)
USER_ID (FK)
MESSAGE
DATE_SENT
STUDENT
STUD_ID (PK)
USER_ID (FK)
GRADE_LEVEL
PARENT_ID (FK)
TEACHER
TEACHER_ID (PK)
USER_ID (FK)
SUBJECT
PARENT
PARENT_ID (PK)
USER_ID (FK)
RELATIONSHIP
MARKS
MARK_ID (PK)
STUD_ID (FK)
SUBJECT
TOTAL_SCORE
HOMEWORK
HWORK_ID (PK)
TEACHER_ID (FK)
TITLE
DUE_DATE
ANNOUNCEMENT
ANNOUNC_ID (PK)
TEACHER_ID (FK)
TITLE
MESSAGE
ATTENDANCE
ATTND_ID
STUD_ID (FK)
DATE
STATUS
HWOR-SUBMISSIONS
SUB_ID (PK)
HWORK_ID (FK)
STUD_ID (FK)
SCORE
AI_PREIDICTIONS
PRED_ID |(PK)
STUD_ID (FK)
RISK_LEVEL
PRED_DATE

The ERD represents the database structure of the School Digital Platform. It shows the main entities, their attributes, and the relationships between them. The diagram supports academic records management, communication features, and AI-based student performance prediction.

FILES

The School Digital Platform will maintain several types of files required for system operation and data storage.

Application Files

These include frontend files, backend files, mobile application files, and AI model files used to run the system.

Database Files

These include database schema files, backup files, and exported reports.

Configuration Files

These include system configuration files, database connection settings, API configuration files, and notification service configuration files.

User Files

These include uploaded homework files, study materials, announcements, and generated reports.

File Formats

The system will use common file formats such as:

- .html, .css, .js (frontend)
- .py (AI and backend processing)
- .sql (database)
- .json (data exchange)
- .pdf and .docx (documents)
- .csv (data export)

All files will be stored securely on the server or cloud storage and access will be controlled through authentication and authorization mechanisms.

SYSTEM PARAMETERS

System parameters are configuration settings that control how the system operates.

Functional Parameters

- Maximum login attempts

- Homework submission deadlines
- Notification intervals
- User role permissions

AI Parameters

- Logistic Regression model parameters
- Risk classification thresholds (Low, Medium, High)
- Minimum data required for prediction
- Model retraining interval

Security Parameters

- Password encryption
- Session timeout
- Access control levels
- Data encryption settings

Performance Parameters

- Maximum number of concurrent users
- Database response time
- API response time
- File upload size limit

Network Parameters

- HTTPS communication
- API endpoint configuration
- Push notification service settings

These system parameters help ensure the system operates efficiently, securely, and reliably.

6. SYSTEM INTERFACES

6.1 USER INTERFACE

The User Interface is the interface through which users interact with the system. The system will provide a web-based interface and a mobile application interface. Users such as students, teachers, parents, and administrators will log into the system to access announcements, homework, academic records, and notifications. The interface will be designed to be user-friendly and accessible on both computers and mobile devices.

6.2 Database Interface

The Database Interface allows the backend system to communicate with the central database. The backend will send queries to the database to store, update, and retrieve data such as student marks, attendance records, homework, and AI prediction results. The database interface will use SQL queries to manage data and ensure data integrity and security.

6.3 AI Module Interface

The AI Module Interface allows the backend system to communicate with the Artificial Intelligence module. The backend will send student data such as marks, attendance, and homework completion to the AI module. The AI module will process this data using a Logistic Regression algorithm and return prediction results such as pass/fail prediction and risk level classification. The results will then be stored in the database and displayed on the teacher dashboard.

6.4 Email Notification Interface

The Email Notification Interface is used to send alerts and notifications to teachers and parents. When the AI system identifies a student as High Risk, the system will automatically send an email notification to the teacher and parent. The system will use an email service such as SMTP to send email alerts.

6.5 Mobile Application Interface

The Mobile Application Interface allows the mobile app to communicate with the backend server through APIs. The mobile app will send user requests to the backend and receive responses such as announcements, homework details, and notifications.

7. SYSTEM PERFORMANCE

The system should:

- Support multiple users at the same time
- Generate AI predictions within a few seconds
- Load pages within 3 seconds
- Store and process large amounts of student data securely

8. GLOSSARY / TERMINOLOGY

Term	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
Database	System used to store data
Logistic Regression	Machine learning algorithm used for prediction
Risk Classification	Categorizing students into risk levels

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